

DATASCI 151 - Introduction to Statistical Computing II

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Course description

Welcome to [DATASCI 151](#)! This Maymester course introduces data analysis and statistical computing with Python and SQL. It is designed for beginners, so you do not need any programming experience to take it.

We will cover the basics of Python programming, Jupyter Notebooks for reproducible research, data wrangling and merging with SQL, and data visualisation.

You will work with real datasets and problems throughout the course and use these tools to answer questions with data. The course builds your technical skills and your critical thinking, which you will need for any data analysis you do later on.

Learning objectives

By the end of this course, you will be able to:

1. Write Python functions and do mathematical operations
2. Clean and manipulate data with Python libraries such as pandas
3. Merge and manage databases with SQL
4. Create visualisations to communicate what you find in the data
5. Use Jupyter Notebooks for reproducible research
6. Solve data analysis and statistical computing problems on your own

Prerequisites

None! Everyone is welcome, whatever your experience with programming or data analysis. If you are not sure whether the course is right for you, just send me an email and I will be happy to help.

Materials

The lecture notes cover everything you need for the course. If you would like to learn more about Python and SQL, all the resources below are free online (the Coursera and edX courses are free if you choose

the audit option).

Suggested books

- [Python for Data Analysis](#) by Wes McKinney
- [Elements of Data Science](#) by Allen Downey
- [Automate the Boring Stuff with Python](#) by Al Sweigart
- [Python for Everybody](#) by Charles Severance
- [Think Python](#) by Allen Downey
- [Python Data Science Handbook](#) by Jake VanderPlas
- [Select Star SQL](#) by Zi Chong Kao

Online courses

- [Coursera: Python for Everybody Specialisation](#)
- [edX: Python Basics for Data Science](#)
- [Software Carpentry: Plotting and Programming in Python](#)
- [Kaggle Learn: Intro to SQL](#)
- [Software Carpentry: Databases and SQL](#)
- [Coursera: SQL for Data Science](#)

Additional resources

- [Python Documentation](#)
- [pandas Documentation](#)
- [SQLite Documentation](#)
- [SQLBolt](#)
- [DataLemur for SQL interview practice](#)
- [GitHub Docs](#)

Course information

We will meet online, Monday to Friday, from 11:00am to 12:20pm EDT. Please review the materials before each class. You can find all course information in our GitHub repository at <https://github.com/danilofreire/datasci151-summer>.

I plan to follow the schedule, but I also want to adapt it to your pace and learning style, so the syllabus may change during the course. I will announce any changes in class and on Canvas. Please also check [the course repository](#) regularly for updates.

Software

We will mainly use [Python](#), a free, general-purpose programming language that is widely used in data science and machine learning. I recommend the [Anaconda distribution](#), which already includes many of the libraries we need, such as [pandas](#), [NumPy](#) and [Jupyter](#).

You can use any text editor you like, but I recommend [VS Code](#) with the [Python extension](#). [PyCharm](#) is another excellent option. Feeling adventurous? Try [Neovim](#) with the [coc-pyright](#) plugin (that is, if [you can exit the editor](#)). :)

We will also use [Jupyter Notebooks](#) in class. Jupyter comes with Anaconda, but please also install the [Jupyter extension for VS Code](#). We will have a hands-on session on how to use it.

I have written [a series of tutorials](#) to help you install Anaconda, VS Code and SQLite, use Jupyter Notebooks and conda environments, and open a free GitHub educational account. Please work through them as soon as possible, so you have all the tools ready when we start.

Office hours

I am flexible with office hours, but email works best. Feel free to write to me at any time at danilo.freire@emory.edu. I usually reply within a few hours.

Academic integrity

Every member of the Emory community is responsible for being honest in all academic work. The [Honor Code of Emory College](#) expects every member of the University to follow the highest ethical standards, and also to refuse to tolerate actions by others that would harm the institution's reputation. Academic misconduct includes any action or inaction that violates the integrity and honesty of the academic community. I will refer any suspected case to the Emory Honor Council.

Artificial intelligence

You may use AI tools for the assignments, as long as they support your own thinking and do not replace it. **AI tools are not allowed during quizzes**, because quizzes are individual tests of your understanding. If you use AI for an assignment, please check the output carefully, as you are responsible for any errors in what you submit. Cite all your sources, including AI tools, and add a short note at the end of any document where you used AI.

Accessibility services

I want every student to succeed, and I am happy to provide accommodations. If you have a medical or health condition that may affect your work, please visit the [Department of Accessibility Services \(DAS\)](#) to learn about the accommodations available. Please send me your DAS Accommodation Letter at the start of the course, or as soon as you receive it.

DAS accommodations such as extra time apply only to quizzes. They do not apply to assignments, because I release assignments in advance and you can work on them at your own pace. If you are an athlete or have other commitments, please tell me about any scheduling conflicts early in the course. I will do my best to accommodate you, although I cannot promise to meet every request. Any questions? Just let me know.

English language learners

Emory welcomes students from across the country and around the world, and international and multilingual students bring perspectives that make our whole community richer. To support multilingual learners, Emory offers language and culture workshops and individual appointments. For more information, please contact the ELLP Specialists at <https://writingcenter.emory.edu>. Your level of English will never be penalised in this course.

Assignments and grading

Assignments (50%). You will complete five assignments during the course. They give you hands-on practice with the lecture material, and each one has conceptual questions and practical exercises. Please complete them individually and acknowledge all your sources, including textbooks, articles and AI tools.

Late submissions automatically receive half credit. Please submit your assignments on Canvas as HTML files, Jupyter notebooks (.ipynb) or PDFs by midnight on the due date.

Quizzes (50%). You will take three in-class quizzes, all on Fridays. Each quiz covers that week's lectures and asks you to apply the concepts to new problems. Quizzes are open-book and open-notes, but **AI tools are not allowed**. They are also individual, so please do not discuss the questions with your classmates.

Grading scale

Your final grade will follow the ranges below, rounded up to the nearest point:

Grade	A	A-	B+	B	B-	C	D	F
Range	91–100	86–90	81–85	76–80	71–75	66–70	60–65	<60

Course outline and suggested readings

The lecture notes cover all the essential material. As I mentioned above, the schedule may change, so please check our [GitHub repository](#) regularly for updates.

Each lecture folder has an HTML file or a Jupyter notebook (.ipynb) with code examples and explanations, plus any datasets or other files we use in class. The suggested readings are all free online.

- Tuesday, May 11: Lecture 01: Welcome to DATASCI 151 - Introduction and Course Overview
 - [Lecture slides](#)
 - Please make sure to install the necessary software for the course by following the [Course Tutorials: How to Install Anaconda, Jupyter, VSCode, and Open a Free Educational Account on GitHub](#)
 - Suggested readings:
 - * Severance, C. [Why should you learn to write programs?](#) Chapter 1 of *Python for Everybody*.
 - * Software Carpentry. [Automated version control](#). In *Version Control with Git*.
 - * GitHub Docs. [Hello World](#).
 - * Visual Studio Code Docs. [Jupyter Notebooks in VS Code](#).
- Wednesday, May 12: Lecture 02: Jupyter Notebooks, Packages, Variables, and Lists
 - [Lecture slides](#)
 - [Lecture notebook](#)
 - Suggested readings:
 - * conda Docs. [Managing environments](#).
 - * Downey, A. [Variables and statements](#). Chapter 2 of *Think Python*, 3rd edition.
 - * Software Carpentry. [Lists](#). In *Plotting and Programming in Python*.
 - * Matplotlib Docs. [Quick start guide](#).
- Thursday, May 13: Lecture 03: Maths Operations, Arrays, and Boolean Logic
 - [Lecture slides](#)

- [Lecture notebook](#)
- [Kahoot quiz](#). Kahoot quizzes are optional and not graded. We will do them together in class, and they are a fun way to review the lecture :)
- Suggested readings:
 - * NumPy Docs. [NumPy: the absolute basics for beginners](#).
 - * Downey, A. [Lists and arrays](#). Chapter 3 of *Elements of Data Science*.
 - * VanderPlas, J. [Aggregations: min, max, and everything in between](#). In *Python Data Science Handbook*.
 - * VanderPlas, J. [Comparisons, masks, and Boolean logic](#). In *Python Data Science Handbook*.
- Friday, May 14: [Lecture 04: Quiz 01 - Based on Lectures 01-03](#). The repository with the questions will be available on the day of the quiz.
 - [Assignment 01 due](#).
- Monday, May 17: Lecture 05: Conditional Statements
 - [Lecture slides](#)
 - [Lecture notebook](#)
 - [Kahoot quiz](#).
 - Suggested readings:
 - * Severance, C. [Conditional execution](#). Chapter 3 of *Python for Everybody*.
 - * Downey, A. [Conditionals and recursion](#). Chapter 5 of *Think Python*, 3rd edition.
 - * Python Docs. [More on lists](#). In *The Python Tutorial*.
- Tuesday, May 18: Lecture 06: For and While Loops
 - [Lecture slides](#)
 - [Lecture notebook](#)
 - [Kahoot quiz](#).
 - Suggested readings:
 - * Severance, C. [Iteration](#). Chapter 5 of *Python for Everybody*.

- * Software Carpentry. [For loops](#). In *Plotting and Programming in Python*.
- * Python Docs. [for statements and the range\(\) function](#). In *The Python Tutorial*.
- Wednesday, May 19: Lecture 07: Functions
 - [Lecture slides](#)
 - [Lecture notebook](#)
 - [Kahoot quiz](#).
 - [Assignment 02 due](#).
 - Suggested readings:
 - * Downey, A. [Functions](#). Chapter 3 of *Think Python*, 3rd edition.
 - * Software Carpentry. [Writing functions](#) and [Variable scope](#). In *Plotting and Programming in Python*.
 - * Python Docs. [Lambda expressions](#). In *The Python Tutorial*.
- Thursday, May 20: Lecture 08: Data Wrangling with Pandas
 - [Lecture slides](#)
 - [Lecture notebook](#)
 - [Kahoot quiz](#).
 - Suggested readings:
 - * McKinney, W. [Data cleaning and preparation](#), sections 7.1 and 7.2. Chapter 7 of *Python for Data Analysis*, 3rd edition.
 - * pandas Docs. [Working with missing data](#).
 - * Python Docs. [Modules](#). In *The Python Tutorial*.
- Friday, May 21: [Lecture 09: Quiz 02 - Based on Lectures 05-08](#). As with Quiz 01, the repository will be available on the day of the quiz.
 - [Assignment 03 due](#).
- Monday, May 24: Lecture 10: Subsetting and Filtering Data
 - [Lecture slides](#)

- [Lecture notebook](#)
- [Kahoot quiz](#).
- Suggested readings:
 - * pandas Docs. [How do I select a subset of a DataFrame?](#)
 - * McKinney, W. [Getting started with pandas](#), section 5.2. Chapter 5 of *Python for Data Analysis*, 3rd edition.
 - * pandas Docs. [How do I create plots in pandas?](#)
 - * Visual Studio Code Docs. [Getting started with Data Wrangler in VS Code](#).
- Tuesday, May 25: Lecture 11: Introduction to SQL
 - [Lecture slides](#)
 - [Lecture notebook](#)
 - [Kahoot quiz](#).
 - Suggested readings:
 - * SQLBolt. [Interactive SQL lessons](#), lessons 1-4 and 10-11.
 - * Software Carpentry. [Selecting data, Filtering, and Aggregation](#). In *Databases and SQL*.
 - * Kao, Z. C. [Select Star SQL](#).
- Wednesday, May 26: Lecture 12: SQL in Python
 - [Lecture slides](#)
 - [Lecture notebook](#)
 - [Kahoot quiz](#).
 - [Assignment 04 due](#).
 - Suggested readings:
 - * Python Docs. [sqlite3 tutorial](#).
 - * Software Carpentry. [Missing data and Programming with databases: Python](#). In *Databases and SQL*.
 - * SQLite Tutorial. [CASE and Window functions](#).
 - * pandas Docs. [Comparison with SQL](#).

- Thursday, May 27: Lecture 13: Merge Tables in SQL
 - [Lecture slides](#)
 - [Lecture notebook](#)
 - [Kahoot quiz](#).
 - Suggested readings:
 - * SQLBolt. [Multi-table queries with JOINS and OUTER JOINS](#).
 - * Software Carpentry. [Combining data](#). In *Databases and SQL*.
 - * SQLite Tutorial. [Self-join and UNION](#).
 - * McKinney, W. [Data wrangling: join, combine, and reshape](#), section 8.2. Chapter 8 of *Python for Data Analysis*, 3rd edition.
- Friday, May 28: [Lecture 14: Quiz 03 - Based on Lectures 10-13](#).
 - [Assignment 05 due](#).